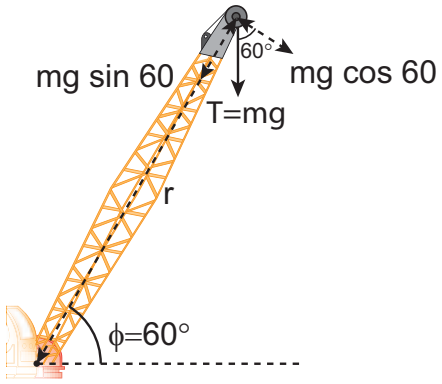
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$$\tau = r(F\perp)$$
$$\tau = r\,mg\,\cos\phi$$
$$\tau = 20\times 20000\times\cos 60^{\circ}$$
$$= 20\times 20000\times\frac{1}{2}$$
$$= 200000\ \text{Nm}$$
$$\tau = 2\times 10^{5}\ \text{Nm}$$

All the three methods, give the same answer.

## 5.2.2 Torque about an Axis

In the earlier sections, we have dealt with the torque about a point. In this section we will deal with the torque about an axis. Let us consider a rigid body capable of rotating about an axis AB as shown in Figure 5.8. Let the force F act at a point P on the rigid body. *The force F may not be on the plane ABP*. We can take the origin O at any random point on the axis AB.

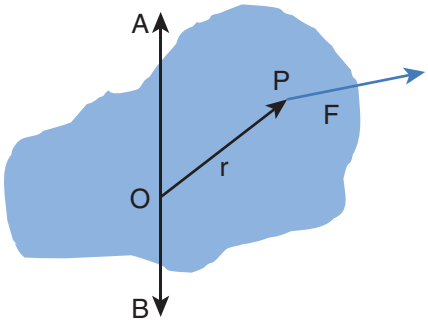



**Figure 5.8.** Torque about an axis



Tamil Nadu is known for creative and innovative traditional games played by children. One such very popular game is "silli" (சில்லி) or "sillukodu" (சில்லுக்கோடு). There is a rectangular area which is further partitioned as seen in the Figure. One has to hop through the rectangles. While doing so, children lean on one side.

**What is the reason?**
Naturally the body takes this position to balance the gravitational force (mg) and Normal force (N) acting on the body and to nullify the torque. Failing which, both these forces act along different lines leading to a net torque which makes one to fall.